

1 Calculus & Trig Rules

$f(t)$	Deriv $f'(t)$	$f(t)$	Integral $\int f(t)dt$
t^n	nt^{n-1}	t^n	$\frac{t^{n+1}}{n+1} \ (n \neq -1)$
e^{at}	ae^{at}	e^{at}	$\frac{1}{a}e^{at}$
$\ln(at)$	$1/t$	$1/t$	$\ln t $
$\sin(at)$	$a \cos(at)$	$\sin(at)$	$-\frac{1}{a} \cos(at)$
$\cos(at)$	$-a \sin(at)$	$\cos(at)$	$\frac{1}{a} \sin(at)$
uv	$u'v + uv'$	Parts	$\int u dv = uv - \int v du$
u/v	$\frac{u'v - uv'}{v^2}$		

Useful Identities:

$$\begin{aligned}\cos^2 \theta &= \frac{1}{2}(1 + \cos 2\theta), \quad \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) \\ \cos \alpha \cos \beta &= \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)] \\ \sin \alpha \sin \beta &= \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]\end{aligned}$$

2 Complex Arithmetic & Exponentials

$$z = x + jy = re^{j\theta} = r\angle\theta$$

Convert: $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}(y/x)$, $x = r \cos \theta$

Euler: $e^{\pm j\theta} = \cos \theta \pm j \sin \theta$

Inv. Euler: $\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$, $\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$

Phasors: $A \cos(\omega t + \phi) = \text{Re}\{Ae^{j\phi}e^{j\omega t}\}$. Amp = $Ae^{j\phi}$.

Time Constant τ : For e^{-at} , $\tau = 1/a$. Signal decays to $\approx 37\%$ of its initial value after one τ .

3 Continuous Transformations

- Shift** by b (left if $b > 0$, right if $b < 0$).
- Scale/Reverse** by a . ($|a| > 1$: compress, $|a| < 1$: expand, $a < 0$: reflect across $t = 0$).

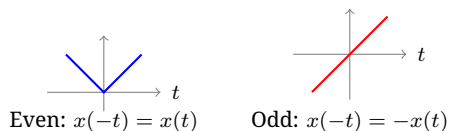
4 Piecewise & Line Equations

To define a graphed signal, find line segments $y = mx + b$:

$$m = \frac{y_2 - y_1}{t_2 - t_1}, \quad b = y_1 - mt_1$$

Rect Pulse $p_T(t)$: $u(t + T) - u(t - T)$ (width $2T$, height 1).

5 Even and Odd Signals



Decomposition: $x(t) = x_e(t) + x_o(t)$

$$x_e(t) = \frac{x(t) + x(-t)}{2}, \quad x_o(t) = \frac{x(t) - x(-t)}{2}$$

6 Periodic Signals

Periodic if $x(t) = x(t + T)$.

Freq: $\omega = \frac{2\pi}{T}$ rad/s.

Sum of Signals: $y(t) = x_1(t) + x_2(t)$ is periodic ONLY if $T_1/T_2 = m/k$ (a rational fraction).

$$\text{Period } T = kT_1 = mT_2$$

7 Unit Step and Dirac Delta

Unit Step $u(t)$: 1 for $t > 0$, 0 for $t < 0$.

Windowing: A signal $x(t)$ strictly between bounds t_1 and t_2 is written: $x(t)[u(t - t_1) - u(t - t_2)]$.

Dirac Delta $\delta(t)$: $\frac{du(t)}{dt} \Rightarrow u(t) = \int_{-\infty}^t \delta(\tau) d\tau$. Area is 1. Zero everywhere except $t = 0$.

• **Equivalence:** $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$

• **Sifting:** $\int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt = x(t_0)$

(Limits MUST enclose t_0 , else result is 0).

• **Time Scaling:** $\delta(at - b) = \frac{1}{|a|}\delta(t - b/a)$

8 Signal Energy and Power

Total Energy E : $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$

Average Power P : $P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$

(For periodic signals: $P = \frac{1}{T} \int_0^T |x(t)|^2 dt$)

Shortcuts: Sinusoid $A \cos(\omega t + \phi) \Rightarrow P = A^2/2$.

DC signal $x(t) = A \Rightarrow P = A^2$.

Classifications:

• **Energy Signal:** $0 < E < \infty \Rightarrow P = 0$.

• **Power Signal:** $0 < P < \infty \Rightarrow E = \infty$.

9 LTI Systems

Linearity: $\mathcal{H}\{ax_1(t) + bx_2(t)\} = ay_1(t) + by_2(t)$

Time-Invariant: $\mathcal{H}\{x(t - t_0)\} = y(t - t_0)$.

Convolution Integral: $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$

Delta Conv: $x(t) * \delta(t - t_0) = x(t - t_0)$

Step Response $s(t)$: $s(t) = \int_{-\infty}^t h(\tau)d\tau$

System Properties & Traps:

• **Causality:** $y(t)$ relies ONLY on present/past inputs. ($x(t + 1)$ or $x(-t)$ for $t > 0$ breaks this).

• **Stability (BIBO):** Bounded input $\rightarrow$ bounded output.

• **Memory:** Memoryless if $y(t)$ depends ONLY on $x(t)$ at the exact same instant t .

• **Invertibility:** Inverse exists if $h(t) * h_{inv}(t) = \delta(t)$.

• **Check Traps:** $y(t) = x(t) + c$ is *Non-Linear*. $y(t) = tx(t)$, $y(t) = x(at)$, and $y(t) = x(t^2)$ are *Time-Varying*.